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2. Study of Network Cables

Aim :-

Study of different types of Network cables.

Procedure :-

a). Understand different types of network cables.

*. Unshielded Twisted Pair (UTP) cable.
*. Shielded Twisted Pair (STP) cable.
*. Coaxial cable.
*. Fiber optic cable.

Cabel Type	Category	Maximum data Transmision	Advantages / Disadvantages	Application / Use
UTP	Category 1	10bps	Advantages :- *. Cheaper in cost *. Easy to install as they have a smaller overall diameter Disadvantages :- *. More prone to (EMI) Electro magnetic interface and noise	10 Base-T Ethernet
	Category 5	Up to 100 Mbps		Fast Ethernet Gigabit Ethernet
	Category 5e	1 Gbps		Fast Ethernet Gigabit Ethernet

Cabel Type	Category	Maximum Data Transmission	Advantages / Disadvantages	Application / Use
STP	Category 6, 6a	10Gbps	Advantages:- *. Shielded *. Faster than UTP *. Less susceptible to noise and interface. Disadvantages:- *. Expensive *. Greater installation effort	Gigabit ethern 10 G Ethernet (55 m) Widely used in data center
SSTP	Category 7	10 Gbps		Gigabit ethern 10 G ethernet (100 m)
Coaxial Cabel	RG-6 RG-56 RG-11	10-100 Mbps	Advantages:- *. High bandwidth *. Immune to interface *. Low Loss bandwidth *. Versatile Disadvantages:- *. Limited distance *. Cost *. Size is bulky	Speed of signal is 500 m Television network, high speed internet connections

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SSTP	Category 7	10Gbps		Gigabit Ethernet, 10G Ethernet (100m)

b). Make your own Ethernet Cross-Over Cabel / Straight cabel :-

Tools required :-

*. Ethernet cabel, CAT5e is certified for gigabit support, but CAT5 cabling work as well just over shorter distances.

*. A crimping tool. This is an all-in one networking tool to push down pins in the plug and strip and cut the shielding off the cabels.

Steps :-

*. To start construction of device begin by threading shield into cabel

*. Next strip approximately 1.5 cm of cabel shielding on both ends. The crimping tool has a round area to complete this task.

*. After you will need to untangle the wires there should be four twisted pairs, Referincing back to the sheet arranging them from top to bottom. One end should be in arrangement A and the other in B.

*. One in order bunch them together. Place them in RJ45 plug without messing up the order.

*. Push the cabel right in. The notch at the end of the plug needs to be just over the cable shielding and if it isint that means you stripped off too much shilding.

*. After the wires are secured insert it into the crimping tool and push down.

*. Lastly repeat for the other end.

Result :-

These are the steps to be followed to securly create our own ethernet cabel either straigth through or cross over.